

Monoalphabetic Substitution Cipher and Cryptanalysis

1. Objective

To implement a monoalphabetic substitution cipher, generate ciphertext from a given plaintext, perform frequency and pattern analysis, and recover the plaintext by using cryptanalysis techniques.

2. Introduction

A monoalphabetic substitution cipher replaces every plaintext letter with a fixed ciphertext letter. The same substitution is used throughout the message.

For example:

```
Plaintext :  A B C D E ...  
Ciphertext:  Q W E R T ...
```

The substitution key contains 26 letters and each plaintext letter has one corresponding ciphertext letter.

3. Encryption

The selected plaintext from *Modern Cryptography* by Katz and Lindell is encrypted using the substitution key.

The encryption process is:

1. Read the plaintext.
2. For every alphabetic character, find its corresponding letter in the substitution key.
3. Replace the letter with its ciphertext equivalent.
4. Keep spaces and punctuation unchanged.
5. The resulting text is the ciphertext.

4. Frequency Analysis

Frequency analysis counts how many times each ciphertext letter occurs.

The program:

- Counts the frequency of each letter.
- Sorts letters in descending order of frequency.
- Calculates the percentage frequency.
- Identifies the most frequent ciphertext letters.

This helps in guessing common plaintext letters such as **E**, **T**, **A**, and **O**.

5. Word Frequency Analysis

The program counts the occurrence of complete words in the ciphertext.

Special attention is given to:

- One-letter words
- Two-letter words
- Three-letter words
- Frequently repeated words

For example, a repeated one-letter word may possibly represent `a` or `I`.

6. Pattern Analysis

Pattern analysis identifies the repeated-letter structure of words.

For example:

```
HELLO → 01223  
APPLE → 01223  
BANANA → 012121
```

If a ciphertext word has the pattern `01223`, words such as `HELLO` or `APPLE` can be considered as possible candidates.

This helps narrow down possible substitutions.

7. Iterative Cryptanalysis

The plaintext is recovered step by step.

The process followed is:

1. Study the letter frequencies.
2. Find repeated words.
3. Analyze word patterns.
4. Propose possible letter substitutions.
5. Apply the substitutions to the ciphertext.
6. Examine the resulting partial plaintext.
7. Reject incorrect guesses.
8. Continue until meaningful plaintext is obtained.

The `display_partial_plaintext()` function is used to observe the plaintext after each guessed substitution.

8. Recovered Substitution Key

After successful cryptanalysis, the complete mapping between ciphertext and plaintext letters is recorded.

Ciphertext	Plaintext
A	–
B	–
C	–
...	...
Z	–

The recovered key represents the final substitution used to decrypt the ciphertext.

9. Verification

The recovered plaintext and substitution key are verified by re-encrypting the plaintext using the recovered key.

If the generated ciphertext is the same as the original ciphertext, the recovered key is considered correct.

10. Conclusion

The monoalphabetic substitution cipher was implemented successfully. Frequency analysis, word frequency, and repeated-letter patterns were used to perform cryptanalysis. By testing substitutions iteratively and examining partial plaintext, the original plaintext and substitution key can be recovered. The solution is finally verified by re-encrypting the recovered plaintext.